Programming Fundamentals

Week 02 - Programming Lab

Introduction

Welcome to your favorite day of the week, which is programming Lab. This week, we will work together to learn and implement new programming concepts including

Skills to be Tested:

- Using special directives to control output on the screen

Let's do some coding.

Task 01(OP): Write and Execute a Program to Print a Game Character. (Pacman)

Sample Output

```
D:\study\programming>c++ "pacman img.cpp" -o pacman.exe

D:\study\programming>pacman.exe
      .::---::..
   .-------------.
 .-----------------.
 ----------------:.
:-----------::.
-----------:.
:-------------:..
 :----------------:.
  .---------------:
    .:---------:.

D:\study\programming>
```

Task 02(CP): Write and Execute a Program to Print a Game Maze.

Sample Output

Skill: Using special directives to control output on the screen

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```
D:\PF codes>c++ maze.cpp -o maze.exe

D:\PF codes>maze.exe
######################################################################
||.. ..................................................  ......  ||
||.. %%%%%%%%%%%%%%%%%%%%          ...      %%%%%%%%%%%%%%%%%%%%  |%|..   %%%%%  ||
||..         |%|    |%|      |%|...      |%|          |%|  |%|..    |%|  ||
||..         |%|    |%|      |%|...      |%|          |%|  |%|..    |%|  ||
||..         %%%%%%%%%%  . . |%|...      %%%%%%%%%%%%%%%%%%      ..   %%%%.  ||
||..         |%|        . . |%|...                ................ |%| ..      .  ||
||..         %%%%%%%%%%%%%. . |%|...     %%%%%%%%%%%%%%%%%%    |%| ..   %%%%.  ||
||..                 |%|.               |%|......        |%| ..    |%|.  ||
||..        .......... |%|.         P    |%|......|%|          ..    |%|.  ||
||..|%|  |%|%%%%|%|.|%|. |%|                .....|%|         ..|%| |%|.  ||
||..|%|  |%|    |%|..     %%%%%%%%%%%%%%%%  .....|%|          .|%|.      ||
||..|%|  |%|    |%|..              ...|%|      %%%%%%%         .|%|.      ||
||..|%|              .              ...|%|                |%| ..|%|.      ||
||..|%|  %%%%%%%%%%%%%%%%%%          ...|%|%%%%%%%%%%%%%%    |%| ..|%|%%%%%%  ||
||.................................................  |%| ..........  ||
||    ..............................................          .......  ||
||..|%|  |%|    |%|..     %%%%%%%%%%%%%%%%  .....|%|     |%| ..|%|.      ||
||..|%|  |%|    |%|..              ...|%|      %%%%%%%    |%| ..|%|.      ||
||..|%|              .      G       ...|%|                |%| ..|%|.      ||
||..|%|  %%%%%%%%%%%%%%%%%%          ...|%|%%%%%%%%%%%%%%    |%| ..|%|%%%%%%  ||
||.................................................  |%| ..........  ||
||   ..............................................          .......  ||
######################################################################

D:\PF codes>
```

Task 03(OP): Write and Execute a Program to Print a Game Character. (Space Craft)

Sample Output

```
      /\
   __/~~\__
 /   |  |   \
=====.  .=====
     ||||
```

Task 04(CP): Write and Execute a Program to Print a Big Alphabet.

- **A**

Skill: Using special directives to control output on the screen

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- **K**
- **Z**
- **K**
- **S**

Sample Output:

```
 ####
##   ##
######
##   ##
##   ##
##   ##

##   ##
##   ##
####
##
####
##   ##
##   ##

######
     ##
    ##
   ##
  ##
##
######

##   ##
## ##
####
###
####
## ##
##   ##

 ####
##   ##
##
 ####
     ##
##   ##
 ####
```

Task 05(CP): Write a c++ program to Print PACMAN with Big Alphabets.

Sample Output (with Multiple Alphabets)

Skill: Using special directives to control output on the screen

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```
D:\study\programming>c++ "pacman text.cpp" -o text.exe

D:\study\programming>text.exe
### ##     ##      ## ##   ##   ##    ##     ###  ##
 ##  ##     ##    ##   ##   ## ##      ##      ## ##
 ##  ##   ## ##   ##       # ### #   ## ##    # ## #
 ##  ##   ##  ##  ##       ## # ##   ##  ##   ## ##
 ## ##    ## ###  ##       ##   ##   ## ###   ##  ##
 ##       ##  ##  ##   ##  ##   ##   ##  ##   ##  ##
####     ###  ##   ## ##   ##   ##  ###  ##  ###  ##

D:\study\programming>
```

Task 06(CP): **Write a c++ program to Print the Header of your project**

```
-----------------------------------------------------------------------------------------------------------------------
|  ######   #####   ######  ###  ##  ######          #####  #######          #####   #####  ####    #######  #####     |
|   ##  ##  ##  ##    ##    #### ##    ##           ##  ##  ##  ##         ##  ##  ##  ##  ##      ##  ##  ##  ##    |
|   ##  ##  ##  ##    ##    #######    ##           ##  ##  ##             ##      ##  ##  ##      ##      ##        |
|   #####   ##  ##    ##    ## ####    ##           ##  ##  ####            #####  #######  ##      ####     #####    |
|   ##      ##  ##    ##    ##  ###    ##           ##  ##  ##                 ##  ##  ##  ##      ##          ##    |
|   ##      ##  ##    ##    ##  ##     ##           ##  ##  ##             ##  ##  ##  ##  ##  ##  ##  ##  ##  ##    |
|  ####      #####   ######  ##  ##     ##            #####  ####           #####  ##  ##  ######  #######  #####    |
|                                                                                                                     |
|            #####   ######   ######   ####     ######   #####   #####   ######   ######   #####   ###  ##           |
|           ##  ##   ##  ##   ##  ##   ##        ##   ##  ##  ##  ##     ##       ##   ##  ##  #### ##              |
|           ##  ##   ##  ##   ##  ##   ##        ##   ##      ##  ##     ##       ##   ##  ##  #######              |
|           #######  #####    #####    ##        ##   ##      #######    ##       ##   ##  ##  ## ####              |
|           ##  ##   ##       ##       ##        ##   ##      ##  ##     ##       ##   ##  ##  ##  ###              |
|           ##  ##   ##       ##       ##  ##    ##   ##  ##  ##  ##     ##       ##   ##  ##  ##   ##              |
|           ##  ##  ####     ####      ######   ######   #####  ##  ##     ##     ######   #####  ##   ##              |
-----------------------------------------------------------------------------------------------------------------------
```

Task 07(CP): **Now, think about your own Business Application and Game that you want to develop in this semester. And draw the header of the Business Application and Game and also Game Player on the console.**

Skill: Using special directives to control output on the screen

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Skill: Printing Colorized Output on the Screen

Introduction

So far we have been printing the desired output on the screen in black and white. In this section of the manual, you will learn to use special functions to print colorized output on the screen.

Let's have some fun with coding.

<table>
<tr><td>01.</td><td>system("Color XY")</td><td>The command is used to set the color of text and background.</td></tr>
<tr><td>02.</td><td>
<table>
<tr><th>Color id</th><th>Color</th><th>Color id</th><th>Color</th></tr>
<tr><td>1</td><td>Blue</td><td>9</td><td>Light Blue</td></tr>
<tr><td>2</td><td>Green</td><td>0</td><td>Black</td></tr>
<tr><td>3</td><td>Aqua</td><td>A</td><td>Light Green</td></tr>
<tr><td>4</td><td>Red</td><td>B</td><td>Light Aqua</td></tr>
<tr><td>5</td><td>Purple</td><td>C</td><td>Light Red</td></tr>
<tr><td>6</td><td>Yellow</td><td>D</td><td>Light Purple</td></tr>
<tr><td>7</td><td>White</td><td>E</td><td>Light Yellow</td></tr>
<tr><td>8</td><td>Gray</td><td>F</td><td>Bright White</td></tr>
</table>
</td><td>X: Background Color
Y: Text Color

The different values are used to print the colorized output in the desired manner. For example,
>> system(“Color 46”)

This would set the background color to the corresponding value of 4(Red) and the text color would be 6(Yellow).</td></tr>
<tr><td>03.</td><td>check out this sample code for a better understanding.</td><td>line.cpp - Notepad
File Edit Format View Help
<pre>
#include<iostream>
using namespace std;

main()
{
    system("color 46");
    cout << "This is My Text";
}
</pre></td></tr>
</table>

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05.	**Output:**	G:\Semesters\Programming Fundamentals (Fall 2023)\Week 2\Class Tasks>c++ line.cpp -o line.exe G:\Semesters\Programming Fundamentals (Fall 2023)\Week 2\Class Tasks>line.exe This is My Text G:\Semesters\Programming Fundamentals (Fall 2023)\Week 2\Class Tasks>

Congratulations, you have just added another skill to your skill set.

Task 05(OP): Write a program to print the Header on a Business Application with colorized Background. (For example UNIVERSITY ADMISSION MANAGEMENT SYSTEM)

```
D:\PF codes>c++ TITLE.cpp -o title.exe

D:\PF codes>title.exe
# # ### ### # # ### ##   ## ### ### # #      #  ##  # # ###  ##  ## ###  #  ###
# # # #  #  # # #   # # #    #   #  # #     # # # # ###  #  #   #    #  # # # #
# # # #  #  # # ##  ##   #   #   #   #     ### # # ###  #   #   #   #  # # # #
# # # #  #  # # #   # #   #  #   #   #     # # # # # #  #   #  #  #  # # # # #
### # # ###  #  ### # # ##  ###  #   #     # # ##  # # ### ##  ##  ###  #  # #
# #  #  ###  #   ## ### # # ### ### ###     ## # #  ## ### ### # #
### # # # # # # #   #   ### #   # #  #     #   # # #    #  #   ###
### ### # # ### # # ##  ### ##  # #  #      #   #   #   #  ##  ###
# # # # # # # # # # #   # # #   # #  #       #  #    #  #  #   # #
# # # # # # # #  ## ### # # ### # #  #     ##   #  ##   #  ### # #

D:\PF codes>_
```

Task 06(CP): Write a program to Print the colorized game maze.

Skill: Printing Colorized Output on the Screen

Programming Fundamentals

Week 02 - Programming Lab

```
D:\PF codes>c++ maze.cpp -o color.exe

D:\PF codes>color.exe
#######################################################################
||.. ..................................................... ......  ||
||.. %%%%%%%%%%%%%%%%        ...     %%%%%%%%%%%%%%  |%|..   %%%%  ||
||..        |%|   |%|     |%|...     |%|        |%|  |%|..    |%|  ||
||..        |%|   |%|     |%|...     |%|        |%|  |%|..    |%|  ||
||..       %%%%%%%%  . .  |%|...     %%%%%%%%%%%%%%     ..   %%%%. ||
||..        |%|       . .  |%|...    ...............  |%| ..       . ||
||..       %%%%%%%%%%. .  |%|...    %%%%%%%%%%%%%    |%| ..   %%%%. ||
||..                |%|.              |%|......       |%| ..    |%|. ||
||..      .......... |%|.        P     |%|......|%|        ..    |%|. ||
||..|%|  |%|%%%|%|.|%|. |%|                .....|%|       ..|%| |%|. ||
||..|%|  |%|    |%|..    %%%%%%%%%%%%%%  ......|%|        .|%|.      ||
||..|%|  |%|    |%|..             ...|%|     %%%%%%        . |%|.    ||
||..|%|             .             ...|%|                 |%| ..|%|. ||
||..|%|  %%%%%%%%%%%%%%%        ...|%|%%%%%%%%%%%     |%| ..|%|%%%%% ||
||.................................................    |%| .......... ||
||   ...............................................          ....... ||
||..|%|  |%|    |%|..    %%%%%%%%%%%%%%  ......|%|     |%| ..|%|. ||
||..|%|  |%|    |%|..             ...|%|     %%%%%%    |%| ..|%|. ||
||..|%|             .     G       ...|%|                 |%| ..|%|. ||
||..|%|  %%%%%%%%%%%%%%%        ...|%|%%%%%%%%%%%     |%| ..|%|%%%%% ||
||.................................................    |%| .......... ||
||   ...............................................          ....... ||
#######################################################################

D:\PF codes>
```

Task 07(CP): Write a Program to print the colorized game character.

Skill: Printing Colorized Output on the Screen

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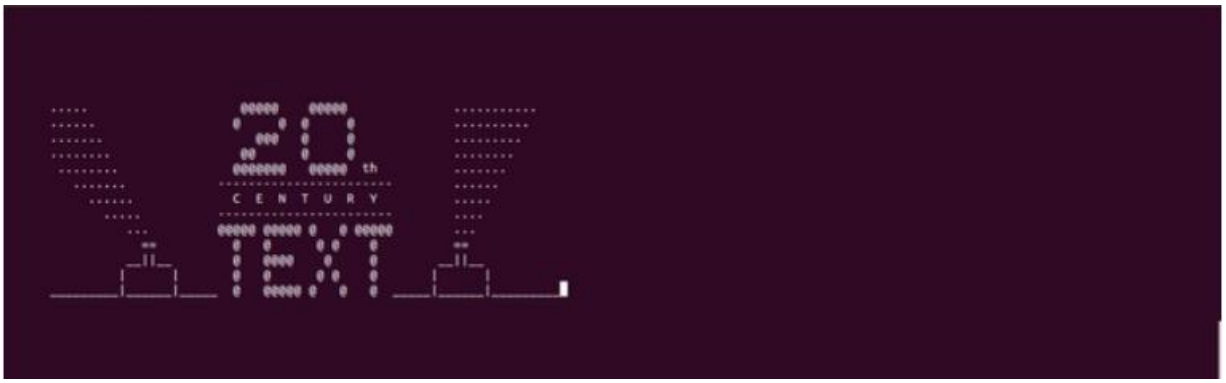
```
D:\PF codes>c++ pacman.cpp -o pac.exe

D:\PF codes>pac.exe
                ..:::::::::..
            .::::::::::::::::.
          .::::::::::::::::::::::.
         ::::::::::::::::::::::::::
       ::::::::::::::::::::::::..
      .::::::::::::::::::..
      .::::::::::::::
      .::::::::::::::::::.
       ::::::::::::::::::::::.
        ::::::::::::::::::::::::::.
         ::::::::::::::::::::::::.
          .::::::::::::::::::::.
             ..::::::::::::..

D:\PF codes>
```

Task 08(CP): Write a c++ program to print the following



Task 09(CP): Write a c++ program to print the following

Skill: Printing Colorized Output on the Screen

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```
-------------------------
        o   ^__^
         o  (oo)\_______
            (__)\       )\/\
                ||----w |
                ||     ||
```

Task 10(CP): Write a c++ program to print the following

```
    8888888888  888    88888
   88     88   88 88   88  88
    8888  88  88   88  88888
       88 88 888888888 88   88
88888888  88 88     88 88    888888

88  88  88   888    88888    888888
88  88  88  88 88   88  88  88
88 8888 88 88   88  88888    8888
 888  888 888888888 88   88     88
  88  88  88     88 88    8888888
```

Task 11(CP): Write a c++ program to print a Car/Truck/ or any Complex vehicle.

Task 12(CP): Write a c++ program to print the following

Skill: Printing Colorized Output on the Screen

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```
          *     *     *
            *     *     *
             *     *     *
            *     *     *
           *     *     *
            *     *     *
              *     *     *
*************************************
 *                                 *
  *                               *
    *                           *
      *                       *
        *********************

 #WELCOME TO F()()!)Y APPLICATITION#
```

Task 13(CP):

```
+--^----------,--------,-----,--------^-,
 | |||||||||   '-------'    |         O
 '+-------------------------^----------|
   ',_________,_________,______________'
    / XXXXXX /'|       /'
   / XXXXXX /  |      /'
  / XXXXXX /'-------'
 / XXXXXX /
/ XXXXXX /
(_______(
 '------'
```

Task 14(CP):

```
';-.          ___'
 '.'!_...._/'.-''
   !       /      ,
   /()   () !    .' '-._
  |)  .    ()!  /   _.'
   |  -'-      ,; '. <
   ;.__     ,;|   > |
  / ,    / ,  |.-'.-'
 (_/    (_/ ,;|.<'
   !     ,     ;-'
    >   !    /
   (_,-''> .'
     (_,'
```

Skill: Printing Colorized Output on the Screen

Programming Fundamentals

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Task 15(CP):

```
                    ___................_
            __.. ' _'.]]- .'-._
 ______.-'        (_) |      ]]           ' ]]'-. _
/_       --------------'-------]]---....______]]__'.'  -..___
| T      _.----._           Xxx|x...           |         _.._'--. _
| |    .' ..--.. '.         XXX|XXXXXXXXXxx==  |       .'.---..'.     -._
]_j   /  /  __  )  )        XXX|XXXXXXXXXXX==  |      / /  __   ) )        '-.
 _|  |  |  /  )  |  |       XXX|'            |     | |  /  )   | |          |
|__)_|  |  (__/  |  |__________|_______________|_____| |  (__/   | |__________J
     '')(      / ./__________________________________( (       / /___________)
        '.'----'.'   dp                                '.'----'.'
          ''                                           ''
```

Good Luck and Best Wishes !!

Happy Coding ahead :)

Skill: Printing Colorized Output on the Screen